

LESSON 7-6 • SESSION 2 • PRACTICE • FORM B

Comparing Distances on the Plane

Learning target – I can decide whether to add or subtract to find a distance, then compare two distances. · 6.NS.C.8

REMEMBER

- Find **each** distance first. Then compare the two numbers.
- Changing coordinates with **opposite signs**: add. **Same sign**: subtract.
- With a **scale**, change both distances to real units before you compare. A distance is never negative.

WORKED EXAMPLE

Jo walks from $(-3, 10)$ to $(-3, 2)$. Sam walks from $(6, -8)$ to $(-7, -8)$. Each unit is 1 block. Who walks farther?

Step 1 Jo: same x , both positive, so $10 - 2 = 8$ blocks.

Step 2 Sam: same y , opposite signs, so $6 + 7 = 13$ blocks.

Answer Sam walks farther, by $13 - 8 = 5$ blocks.

WORDS distance (*distancia*) · quadrant (*cuadrante*) · compare (*comparar*)

PRACTICE

show your work · label units

1. Add or subtract? Find each distance.

a. $(4, -2)$ to $(4, -13)$ _____

b. $(4, -2)$ to $(4, 8)$ _____

c. $(-3, 2)$ to $(6, 2)$ _____

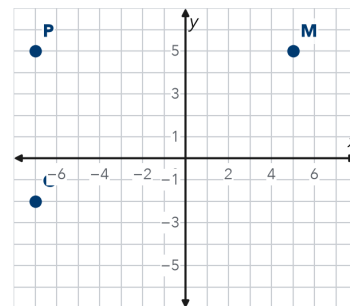
d. $(7, -9)$ to $(7, -1)$ _____

2. **Nature Center Map.** The map shows a pond (P), a meadow (M) and a cabin (C). Each grid unit stands for 10 meters.

a. Pond to Meadow _____ m

b. Pond to Cabin _____ m

c. Which trip is longer? By how many meters? _____



3. **Two Riders.** From $(-8, 4)$, Rosa rides to $(-8, -7)$ and Kofi rides to $(6, 4)$. Each unit is 1 block. Who rides farther, and by how much?

Hint Each ride keeps one coordinate. Check the signs of the other one.

Show your work.

By _____

4. **Partner Point.** Point T is at $(-3, 14)$. Point U is 20 units directly below T. Name point U.

Hint Moving down changes only y : $14 - 20$.

Show your work.

U = _____

5. **Place the Stop.** Point N is at $(12, -4)$. Point M is 18 units directly to the left of N. Name point M.

Hint Moving left changes only x : $12 - 18$.

Show your work.

M = _____

- 6. Two Hikes.** Mei hikes from $(-10, 13)$ to $(-10, 4)$. Andre hikes from $(-3, -14)$ to $(11, -14)$. Each unit is 1 km. Who hikes farther, and by how much?

Show your work.

By _____

- 7. Two Walks.** From $(3.25, 4.75)$, Ana walks to $(3.25, -2.25)$ and Ben walks to $(3.25, 9.75)$. Who walks farther, and by how much?

Hint Decimals work the same way. Check the signs of the y-values.

Show your work.

By _____

- 8. Delivery Routes.** A depot is at $(0, 0)$. Stop 1 is at $(0, 11)$, Stop 2 is at $(0, -8)$ and Stop 3 is at $(0, 5)$.

- Find the distance from the depot to each stop. _____
- Order the stops from closest to farthest. _____
- Find the distance from Stop 1 to Stop 2. _____ units

FIND THE ERROR

explain the mistake, then fix it

- 9. Diego's Work.** To find the distance from $(-6, 2)$ to $(9, 2)$, Diego wrote $9 - 6 = 3$ units. Explain his mistake. Then find the correct distance.

TEST PRACTICE

choose the best answer

- 10.** Two coordinates on opposite sides of zero call for –
A subtraction **B** addition **C** division **D** doubling
- 11.** The distance from $(-7, 0)$ to $(7, 0)$ is –
A 0 units **B** 7 units **C** 14 units **D** 49 units

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

(30, -4) Z	2 R	3 M	19 S	1 X	(-3, -6) E	5 E	(-3, 34) Q	2.5 J	(-6, -4) T
		#3	#4	#5	#6	#7	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____